ML & Data Roles – Job Titles, Core Skills, and Day-to-Day Responsibilities

Job Title	Core Skills	Must-Have Day-to-Day Responsibilities
MLOps Engineer / Senior MLOps Engineer	Python for production ML (APIs, testing, packaging) CI/CD for ML (GitHub Actions, GitLab CI, Jenkins) Docker & Kubernetes for model serving (EKS/GKE/AKS) Cloud ML stack (SageMaker, Vertex AI, Azure ML) Pipeline orchestration (Airflow, Kubeflow, Prefect) Model registry & experiment tracking (MLflow, W&B;) Monitoring & observability (Prometheus, Grafana, logs) Infrastructure as Code (Terraform/CloudFormation)	Build and maintain automated training and deployment pipelines for ML models Containerize models and deploy them to Kubernetes-based serving infrastructure Monitor production models for latency, drift, and performance regressions Troubleshoot failing ML pipelines and production incidents end-to-end Collaborate with data scientists to productionize notebooks into reliable services Manage cloud resources and costs for ML workloads (compute, storage, networking) Implement and maintain model registries, versioning, and promotion workflows Work with security/cloud teams to harden ML infra (IAM, secrets, network policies)
ML Platform Engineer	Strong software engineering (Java/Go/Python, APIs, microservices) Kubernetes & cloud-native infrastructure (autoscaling, service mesh) ML orchestration platforms (Kubeflow, Vertex AI Pipelines, SageMaker Pipelines) Feature stores & model registries (Feast, Databricks, Hopsworks, MLflow) Distributed data processing (Spark, Beam, Flink) Infrastructure as Code (Terraform) & CI/CD for platform services Cloud expertise (AWS/GCP/Azure) including networking & IAM Observability (metrics, logs, tracing) and SLO management	Design and evolve the internal ML platform (training, serving, feature store) Provide self-service tooling and APIs for data scientists and ML engineers Scale platform capacity and reliability to meet latency and availability SLOs Implement observability and incident response processes for ML platform services Onboard teams to standardized pipelines, templates, and SDKs/CLIs Collaborate with security, data, and compliance teams on governance and access control Optimize cloud resource utilization and drive cost efficiencies for ML workloads Plan and execute migrations from legacy ML infra to centralized platform solutions
Machine Learning Engineer	Python and ML libraries (scikit-learn, XGBoost, LightGBM) Deep learning (PyTorch/TensorFlow/Keras) where relevant Statistics and ML fundamentals (evaluation, experimentation) Data wrangling with SQL, Pandas, and Spark Model serving (REST/gRPC services, batch scoring) Basic MLOps concepts (registry, monitoring, CI/CD for models) Cloud ML services (SageMaker, Vertex AI, Azure ML) Collaboration skills with product, data, and security teams	Explore data and engineer features from large-scale datasets/logs Train, evaluate, and tune ML models to hit product and accuracy targets Compare candidate models via robust validation and experiment design Deploy and maintain inference services with appropriate SLAs Monitor model performance and trigger retraining when data or behavior drifts Work with data engineers to define data contracts and feature pipelines Partner with product/security teams to translate requirements into ML solutions Document designs and results, and present findings to technical and non-technical stakeholders

Data Engineer (ML-focused)	SQL and data modeling for analytical and ML use cases Python/Scala for ETL/ELT and pipeline development Batch and streaming processing (Spark, Beam, Flink) Job orchestration (Airflow, Prefect, Dagster) Streaming platforms (Kafka, Kinesis, Pub/Sub) Data warehouses/lakes (BigQuery, Snowflake, Redshift, Lakehouse) Feature store patterns and ML-ready data serving Data quality, lineage, and observability tooling	Design and implement batch and streaming pipelines that supply ML-ready data Model and maintain warehouse/lake schemas optimized for feature computation Implement data quality checks, validation, and monitoring on critical ML tables Build and maintain feature stores or equivalent abstractions for ML teams Integrate multi-source data (logs, telemetry, business data) into unified datasets Collaborate with ML engineers on feature definitions and reproducible training datasets Harden data pipelines with access control, encryption, and audit logging Optimize pipeline performance and cost (partitioning, caching, job scheduling)